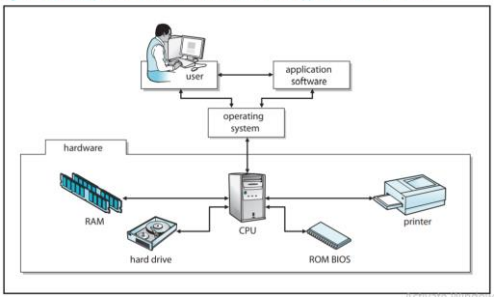
Software Tools

NGUYEN Duc Dung, Ph.D.
Institute of Information Technology, VAST

What is an Operating System?

Figure 9-1, An OS provides an interface between the user, applications, and hardware



2

What is an OS?

"The **program** responsible for **managing** the ***user interface***, system ***resources***, and ***processes***."

"OS is a **software system** that allows ***multiple software*** applications to use a ***single set of hardware*** components at the **same time**."

"OS manages all of the **software** and **hardware** on the computer. Most of the time, there are several different computer programs running at the same time, and they all need to access your computer's **CPU**, **memory**, and **storage**. The OS coordinates all of this to make sure each program gets what it needs."

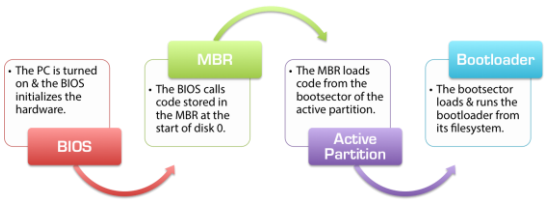
3

Informatics: Course Content

- History of Computing and Computer Architecture *(no practice)*
- Operating System
- **Software Tools for Techies**
- Data Presentation and File Structures
- Networks and the Internet
- Introduction to Programming and Software Engineering

4

BIOS/MBR Boot Process

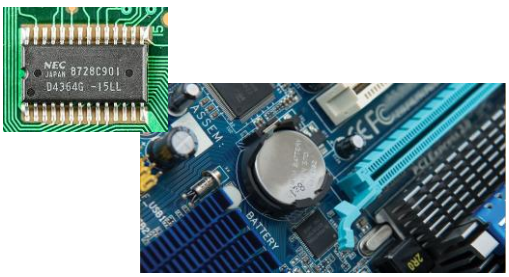


BIOS (Basic Input/Output System) :

- Is **firmware** used to perform hardware initialization
- Provides runtime **services** for OS and programs.
- **Pre-installed** on a personal computer's system board.

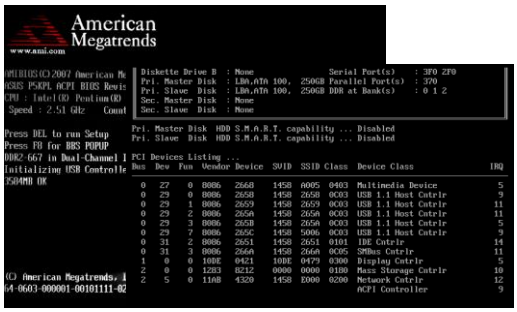
5

BIOS memory and CMOS Battery



6

Power-On Self-Test (POST)

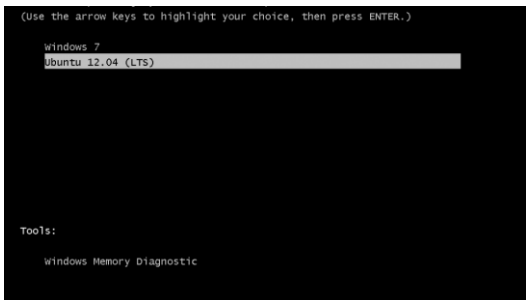


POST Errors

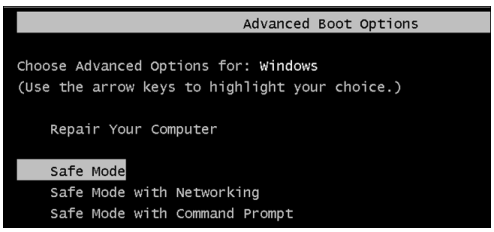
Beeps	Meaning
1	No RAM installed/detected
2	Incomp
3	No goo
4	No goo
5	Proces

Color	Meaning
Red	Bad ROM
Yellow	CPU Exception Before Bootstrap Code is Loaded
Green	Bad Chip RAM or fail of Agnus Chip (check seating of Agnus)
Black	No CPU
White	Expansion passed test successfully
Grey	Turn on
Constant white	Failure of CPU
Violet	Return from InitCode()[5]

Boot Menu



Boot Options

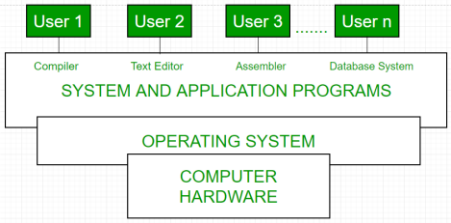


Homework





What is an Operating System?

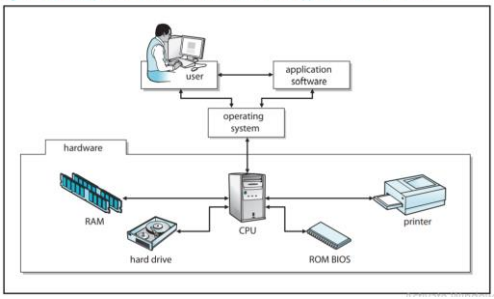


"OS is a software system that allows ***multiple software*** applications to use a ***single set of hardware*** components at the **same time**."

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OS: Basic Functions

Figure 9-1. An OS provides an interface between the user, applications, and hardware



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OS: Basic Functions and Tools

Functions

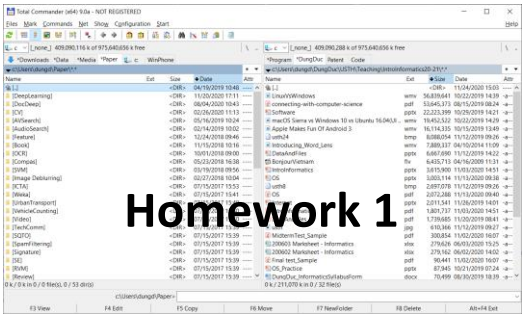
- **System Administration**
  - Resource management
  - Hardware, software
  - Networks
  - Security
- **Information Processing**
  - Text
  - Image
  - Number
  - Multimedia

Tools

- **Windows Settings**
  - User account, group
  - Device/software manager
  - Defenders
  - File manager
- **Software**
  - Notepad, MS Word
  - Paints, Photoshop
  - MS Excel
  - MS PowerPoint

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File Manager: Total Commander



Homework 1

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Text Editors (Homework 2)

WYSIWYG Editors

- **Examples:** MS Word, LibreOffice, GoogleDoc
- **Advantages:**
  - Easy of use
  - Fast for small docs
- **Limitation:**
  - Price
  - Scientific features

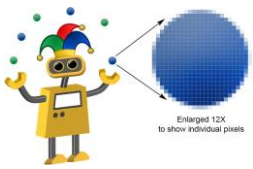
Markup Editors

- **Examples:** Latex, HTML
- **Advantages:**
  - Price
  - Fast for large docs
  - Scientific features
- **Limitations:**
  - Hard of use

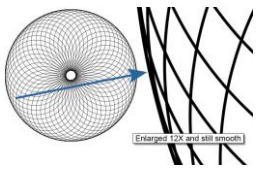
17

Image: Bitmap vs. Vector

Bitmap



Vector



18

Image: Bitmap vs. Vector

Bitmap

- Made of shapes (line, polygons,…)
- Lose quality when the image is resized larger
- Popular
- Ubiquitous format: BMP, JPEG, GIF, PNG, …

Vector

- Made of pixels
- Scalable without losing quality
- Specialized uses
- Specialized format: TTF, EPS,

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Image: Bitmap vs. Vector

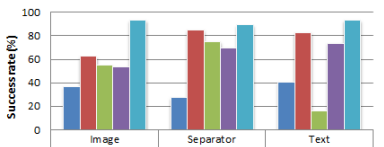
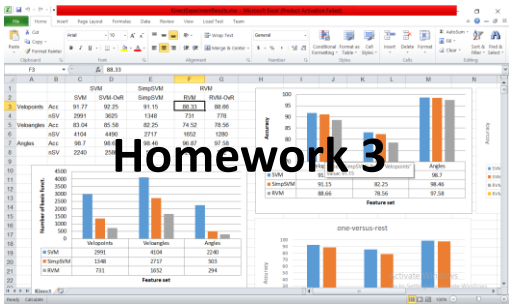


	Image	Separator	Text
DICE	36.46	27.08	40.02
Fraunhofer	61.83	84.51	82.37
REGIM-ENIS	54.42	74.53	15.44
Tesseract	52.95	69.42	73.24
AOSM	92.65	88.75	92.66

			Text
			40.02
Fraunhofer	61.83	84.51	82.37
REGIM-ENIS	54.42	74.53	15.44
Tesseract	52.95	69.42	73.24
AOSM	92.65	88.75	92.66

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Calculation and Graphing on Datasheet



Homework 3

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Practice and Mid-term Test

Basic information processing

(text, image, data sheets, presentation)

- Creating files
- Processing information
- Storing files

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